A CERTIFIED BRACKET OF WIDTH $1.5 \cdot 10^{-8}$ FOR THE L^4 NORM OF THE CYCLIC HILBERT TRANSFORM ON $\mathbb{Z}/31\mathbb{Z}$

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Abstract. Let H be the cyclic Hilbert transform on $\mathbb{Z}/31\mathbb{Z}$, given by the Fourier multiplier $\widehat{Hf}(k) = -i\,\mathrm{sgn}(k)\widehat{f}(k)$, and let C_{31} be its L^4 operator norm on real zero-mean functions. We prove

$$1.5693754353475169 \;<\; C_{31} \;<\; \tfrac{31387509}{20000000} = 1.56937545.$$

The previously recorded bracket was $1.5693 < C_{31} < 1.6453$. The upper bound is a degree-four sum-of-squares certificate. It has three ingredients: a translation-invariant Gram matrix on the 496 quadratic monomials; a cubic multiplier of $\sum_x f(x)$, which restricts the inequality to the zero-mean subspace; and an explicit residual that is controlled by rational interval arithmetic. Positive definiteness is certified by an exact identity $Q - \gamma_0 I = LL^{\mathsf{T}} + R$ with integer data. The lower bound is a rational witness. A standalone checker written only with the Python standard library verifies both bounds in exact rational arithmetic. Numerically, the local maximizer has value $C_{31} = 1.5693754353475169157836\ldots$; it is reflection-symmetric and is a nondegenerate local maximum. The certified bracket shows that no critical value exceeds it by more than $1.5 \cdot 10^{-8}$. Further numerical evidence indicates that the sum-of-squares relaxation is exact.

1. Introduction

On the circle, the sharp L^p norm of the conjugate function operator was found by Pichorides [10], completing work of M. Riesz [11]. On the lattice $\mathbb{Z}$, the corresponding problem for the discrete Hilbert transform was solved by Bañuelos and Kwaśnicki [7]. For finite cyclic groups, the sign multiplier gives a natural finite-dimensional analogue. Its sharp norm is the maximum of a ratio of quartic forms, and it is not known in closed form even for small odd orders. Problem P2730 of TheoremDB [14] asks for the sharp L^4 constant at order 31. It records the bracket $1.5693 < C_{31} < 1.6453$. The lower bound there comes from an interval-certified rational witness near the numerically found local maximum 1.5693754353475171 [13]. The upper bound comes from Riesz–Thorin interpolation.

The main result of this paper narrows the bracket by more than five orders of magnitude.

Theorem 1.1. *Let C_{31} be the supremum of $\|Hf\|_4/\|f\|_4$ over real nonzero f on $\mathbb{Z}/31\mathbb{Z}$ with $\sum_x f(x) = 0$. Then*

$$1.5693754353475169 < C_{31} < U := \frac{31387509}{20000000}.$$

More precisely, every real f on $\mathbb{Z}/31\mathbb{Z}$ with $\sum_x f(x) = 0$ satisfies

$$U^4 \sum_x f(x)^4 - \sum_x (Hf)(x)^4 \;\geq\; 1.83 \cdot 10^{-9} \sum_x f(x)^4. \tag{1}$$

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The proof of the upper bound is a computer-assisted sum-of-squares (SOS) argument [6, 8]. Two features keep the verification short and exact.

First, the Gram matrix is invariant under translations of $\mathbb{Z}/31\mathbb{Z}$. This symmetry reduction [3] is used only to *find* the certificate. The checker itself works with the full 496×496 matrix.

Second, the identity contains the term $\big(\sum_x f(x)\big)\, C(f)$ with a cubic form C. This term vanishes on the zero-mean subspace. Without it, the best degree-four SOS bound is (numerically) the norm of H on all of $\mathbb{R}^{31}$, which is about 1.5698391; this is strictly larger than C_{31}, because the constraint $\sum f = 0$ is active. A working draft [1] certified the full-space bound $C_{31} < 1.574$ with the same translation-invariant Gram construction but without the multiplier. Adding the multiplier is what makes the relaxation tight.

Following the strategy of rationalizing a numerical SOS solution [9, 5], the certificate consists of integers only, and every inequality used in the proof is checked by exact rational comparisons. The sole transcendental input is the kernel $h(j)$, a sum of sines. The checker encloses it from scratch: π by Machin's formula, and the sine by its Taylor series with explicit remainder.

Section 7 records numerical facts that are not part of the proof. The extremizer is symmetric under $x \mapsto 3 - x$, it is a nondegenerate local maximum, and its value agrees with the SOS relaxation to solver accuracy. We also tabulate C_n for other odd orders n.

2. The operator and the quartic problem

Throughout, $n = 31$ and indices are taken modulo n. For real $f = (f_0, \dots, f_{30})$ we use

$$\widehat{f}(k) = \frac{1}{n} \sum_x f_x e^{-2\pi i k x/n}, \qquad \widehat{Hf}(k) = -i \operatorname{sgn}(k) \widehat{f}(k), \quad -15 \le k \le 15.$$

Fourier inversion turns H into a convolution with a real odd kernel:

$$(Hf)_x = \sum_y h(x-y) f_y, \qquad h(j) = \frac{2}{n} \sum_{k=1}^{15} \sin \frac{2\pi k j}{n} = \frac{\cos(\pi j/n) - (-1)^j}{n \sin(\pi j/n)} \quad (j \neq 0), \tag{2}$$

with $h(0) = 0$. As matrices, $H^{\mathsf{T}} = -H$, $H\mathbf{1} = 0$ and $H^{\mathsf{T}} H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^{\mathsf{T}}$. Translations commute with H, and the reflection $f_x \mapsto f_{-x}$ anticommutes with it. Consequently the ratio $\|Hf\|_4 / \|f\|_4$ is invariant under the dihedral group and under nonzero scaling. The normalization of $\|\cdot\|_4$ does not matter because only the ratio enters.

Put

$$F(f) = \sum_x (Hf)_x^4, \qquad G(f) = \sum_x f_x^4, \qquad P_U = U^4 G - F.$$

The zero-mean unit sphere of G is compact, so $C_{31}^4 = \max\{F(f)/G(f) : \sum f = 0,\ f \neq 0\}$ is attained. The upper bound in Theorem 1.1 is the statement that $P_U > 0$ on the zero-mean subspace minus the origin.

3. The certificate and why it implies the theorem

3.1. **Quadratic monomials and the Gram matrix.** For $0 \le d \le 15$ and $t \in \mathbb{Z}/31\mathbb{Z}$ let $z_{d,t}(f) = f_t f_{t+d}$. Because $15 < 31/2$, these $16 \cdot 31 = 496$ products are exactly the distinct quadratic monomials. Let $z(f) \in \mathbb{R}^{496}$ be the vector of all of them. A real array $K = (K_{d,e,s})$ of shape $16 \times 16 \times 31$ defines a matrix

$$Q_{(d,t),(e,u)} = K_{d,e,u-t}. \tag{3}$$

This Q is invariant under simultaneous translation of t and u, and it is symmetric if and only if $K_{d,e,s} = K_{e,d,-s}$. The quartic form $z(f)^{\mathsf{T}} Q z(f)$ is what the certificate uses.

3.2. **The multiplier.** Let $\mathcal{B}$ be a set of representatives of the translation orbits of 3-element multisets in $\mathbb{Z}/31\mathbb{Z}$; there are $\binom{33}{3}/31 = 176$ of them. Given rational numbers $(c_B)_{B\in\mathcal{B}}$, put

$$C(f) = \sum_{B=\{b_1,b_2,b_3\}\in\mathcal{B}} c_B \sum_t f_{t+b_1} f_{t+b_2} f_{t+b_3}.$$

The product $\big(\sum_x f_x\big)C(f)$ is a quartic form that vanishes on the zero-mean subspace.

3.3. **The reduction lemma.**

Lemma 3.1. *Let Q be a real symmetric 496×496 matrix and C a real cubic form. Suppose the following hold.*

(a) *$Q \succeq \mu I$ for some $\mu > 0$.*
(b) *The quartic form $E = P_U - z^{\mathsf{T}}Qz - \big(\sum_x f_x\big)C$ has monomial coefficients e_α with $\delta := \sum_\alpha |e_\alpha| < \mu$.*

Then $P_U(f) \ge (\mu-\delta)G(f)$ for every real f with $\sum_x f_x = 0$. In particular $C_{31} < U$.

Proof. Let $\sum_x f_x = 0$. By (a), $z^{\mathsf{T}}Qz \ge \mu\|z\|^2$. Keeping only the monomials with $d = 0$ gives $\|z\|^2 \ge \sum_t z_{0,t}^2 = G(f)$. For any degree-four monomial $f^\alpha = f_a f_b f_c f_d$, the arithmetic–geometric mean inequality gives $|f^\alpha| \le \frac14(f_a^4+f_b^4+f_c^4+f_d^4) \le \max_j f_j^4 \le G(f)$, and hence $|E(f)| \le \delta G(f)$. The multiplier term vanishes. Therefore $P_U(f) = z^{\mathsf{T}}Qz + E(f) \ge (\mu-\delta)G(f)$. □

Lemma 3.2. *Let Q be symmetric, and suppose $Q - \gamma_0 I = LL^{\mathsf{T}} + R$ with R symmetric. Let $\rho = \max_i \sum_j |R_{ij}|$. Then $Q \succeq (\gamma_0 - \rho)I$.*

Proof. $LL^{\mathsf{T}} \succeq 0$, and for symmetric R we have $\|R\|_2 \le \sqrt{\|R\|_1\|R\|_\infty} = \rho$. Therefore $Q - \gamma_0 I \succeq R \succeq -\rho I$. □

3.4. **The certificate data.** The certificate file `certificate.json` contains integers only. With all scalings stated explicitly, it consists of the following.

- $U = 31387509/20000000$ and $\gamma_0 = 2^{-29}$.
- $K_{d,e,s} \in 2^{-50}\mathbb{Z}$: 7936 integers.
- $c_B \in 2^{-50}\mathbb{Z}$ for the 176 orbit representatives B.
- A lower-triangular $L \in 2^{-45}\mathbb{Z}^{496\times 496}$: 123,256 integers.
- A witness $q \in \mathbb{Z}^{31}$ with $\sum_x q_x = 0$, used for the lower bound.

4. Exact verification

The checker `verify.py` (Section 8) uses only the Python standard library. It never calls the generating code. It performs the steps below and asserts each inequality with exact `Fraction` or integer arithmetic.

4.1. **Enclosure of the kernel.** First, π is enclosed using $\pi = 16\arctan\frac15 - 4\arctan\frac{1}{239}$, with 200 and 80 terms of the alternating series respectively and the first omitted term as the error bound. The midpoint is rounded to the grid $2^{-300}\mathbb{Z}$, and the radius r_π includes the rounding.

Next, for $m = 0,\dots,30$ let $x_m = 2\tilde\pi m/31$, where $\tilde\pi$ is that midpoint. The checker sums 120 terms of the Taylor series of $\sin x_m$ and bounds the remainder by the Lagrange estimate $|x_m|^{241}/241! < 10^{-100}$. It then adds the Lipschitz error $|x - x_m| \le 2r_\pi m/31$ and rounds to $2^{-300}\mathbb{Z}$.

The values $\tilde h(j)$ obtained from (2) are rounded to $2^{-200}\mathbb{Z}$, and the checker proves

$$|\tilde h(j) - h(j)| \le \varepsilon_h < 2^{-200}, \qquad |\tilde h(j)| + \varepsilon_h < 1 \quad (0 \le j \le 30).$$

As an independent sanity check, not used in the proof, the actual deviation from a 70-digit `mpmath` [12] evaluation is $2.8 \cdot 10^{-61}$, while the claimed radius is $6.2 \cdot 10^{-61}$.

4.2. **Positive definiteness.** The checker builds Q from (3) and asserts that it is exactly symmetric. It then forms $R = Q - \gamma_0 I - LL^{\mathsf{T}}$ entrywise with integer arithmetic. To make this exact, every entry is scaled by 2^{90}; the products of L are formed exactly. The checker finds

$$\rho < 8.6 \cdot 10^{-12}, \qquad \mu := \gamma_0 - \rho > 1.854 \cdot 10^{-9},$$

so Lemma 3.2 gives hypothesis (a) of Lemma 3.1.

4.3. **Coefficient comparison.** The checker expands $z^{\mathsf{T}}Qz$ (496^2 terms) and $\big(\sum f\big)C$ ($176 \cdot 31^2$ terms) into monomials. For each of the $\binom{34}{4} = 46{,}376$ monomials f^α it computes, exactly,

$$\tilde{e}_\alpha = U^4[\alpha \text{ is a fourth power}] - \tilde{F}_\alpha - \mathrm{coef}_\alpha(z^{\mathsf{T}}Qz) - \mathrm{coef}_\alpha\big((\textstyle\sum f)C\big),$$

where

$$\tilde{F}_\alpha = \frac{4!}{\prod_j m_j!} \sum_x \prod_j \tilde{h}(x-j)^{m_j}$$

is the coefficient of f^α in F with h replaced by $\tilde{h}$ (m_j is the multiplicity of j in α). No orbit bookkeeping is used. Since $|h|, |\tilde{h}| < 1$, telescoping gives $\big|\prod_{i=1}^4 h_i - \prod_{i=1}^4 \tilde{h}_i\big| \le 4\varepsilon_h$, so $|e_\alpha - \tilde{e}_\alpha| \le \frac{4!}{\prod m_j!} \cdot 31 \cdot 4\varepsilon_h$. The multinomial coefficients sum to 31^4, so in total

$$\delta \le \sum_\alpha |\tilde{e}_\alpha| + 124 \cdot 31^4 \varepsilon_h < 2.05 \cdot 10^{-11} + 10^{-52}.$$

4.4. **Conclusion of the upper bound.** Hence $\mu - \delta > 1.83 \cdot 10^{-9}$, and Lemma 3.1 proves (1) and the upper half of Theorem 1.1. The checker's actual output is reproduced in Table 1.

TABLE 1. Margins reported by the exact checker. Decimal values are rounded outward; the proof uses only the exact rational comparisons.

Quantity	Value
γ_0	$2^{-29} \approx 1.8626 \cdot 10^{-9}$
$\rho = \max_i \sum_j \lvert R_{ij}\rvert$	$8.505 \cdot 10^{-12}$
$\mu = \gamma_0 - \rho$	$1.854141 \cdot 10^{-9}$
$\sum_\alpha \lvert\tilde{e}_\alpha\rvert$	$2.049 \cdot 10^{-11}$
trigonometric error	$7.1 \cdot 10^{-53}$
$\mu - \delta$	$1.833647 \cdot 10^{-9}$

5. The lower bound

The witness $q \in \mathbb{Z}^{31}$ is the 80-digit numerical maximizer (Section 7) scaled by 2^{80} and rounded. The last coordinate is set so that $\sum q_x = 0$ exactly. For each x, the checker forms the interval $(Hq)_x \in \big[m_x - r_x,\ m_x + r_x\big]$ with

$$m_x = \sum_y \tilde{h}(x-y) q_y, \qquad r_x = \varepsilon_h \sum_y |q_y|,$$

asserts $|m_x| > r_x$, and bounds $F(q) \ge \sum_x (|m_x| - r_x)^4$. Comparing with $G(q)$ in exact arithmetic gives

$$F(q) > (1.5693754353475169)^4\, G(q),$$

which proves the lower half of Theorem 1.1.

6. How the certificate was found

This section is not needed to check the proof.

6.1. **The semidefinite program.** Fourier transformation in t block-diagonalizes (3) into 31 Hermitian matrices of size 16×16:

$$B_k(d,e) = \sum_s K_{d,e,s} e^{-2\pi i k s/31}, \qquad B_{31-k} = \overline{B_k}.$$

The eigenvalues of Q are those of $B_0, \dots, B_{30}$. For a prescribed margin γ we solve

$$\min t \quad \text{s.t.} \quad tG - F = z^{\mathsf{T}} Q(K) z + \big(\textstyle\sum f\big) C, \qquad B_k(K) \succeq \gamma I \ (0 \le k \le 15),$$

where the equality is imposed on the 1496 translation orbits of quartic monomials. The variables are K (with the symmetry $K_{d,e,s} = K_{e,d,-s}$), the 176 coefficients c_B, and t. We used CVXPY [2] with the Clarabel interior-point solver [4]; each solve takes about a minute.

With $\gamma = 10^{-8}$ the solver returned $t^{1/4} = 1.56937544913$. The true minimum eigenvalue of the returned Q was $4.76 \cdot 10^{-9}$; the solver's limited accuracy loses about half of the requested margin.

6.2. **Rationalization.** We rounded c and K to $2^{-50}\mathbb{Z}$, symmetrizing K in the process. The orbit equalities were then restored, up to one grid unit, by adjusting a single designated pair $K_{d,e,s} = K_{e,d,-s}$ in each orbit. Target coefficients were computed with 80-digit `mpmath`. Because each entry of K contributes to exactly one orbit, these adjustments do not interfere with each other. The gap $U^4 - t$ lands on the orbit of f_t^4, that is, on the diagonal entries $Q_{(0,t),(0,t)}$, so it only increases Q.

Finally, a floating-point Cholesky factor of $Q - 2^{-29} I$ was rounded to $2^{-45}\mathbb{Z}$. Its rounding error is exactly what the residual R of Lemma 3.2 accounts for.

7. Numerical observations

The statements in this section are numerical and are not claimed as theorems, except Proposition 7.1.

7.1. **The extremizer.** Among 400 BFGS runs from random starts on the zero-mean subspace, 154 converged to the same value, and no other local maximum came within 10^{-9} of it. Up to the dihedral action and scaling, the maximizer satisfies $f(x) = f(3-x)$, has alternating signs, and has a slowly varying envelope.

We ran Newton's method on the stationarity equations restricted to this symmetric subspace, in 80-digit arithmetic. Its limit is stationary on the full zero-mean subspace: the projected gradient has norm about $2 \cdot 10^{-79}$. The value is

$$C_{31} \stackrel{?}{=} 1.56937543534751691578362427973611653862455578221348682 7\ldots$$

This agrees with the value recorded in [13] and with Theorem 1.1. The Hessian of F/G on the zero-mean subspace has 29 negative eigenvalues, between -121.4 and -14.3, and one zero eigenvalue along the scaling direction. So the critical point is a nondegenerate strict local maximum modulo scaling.

Theorem 1.1 shows that no zero-mean critical value exceeds this one by more than $1.5 \cdot 10^{-8}$. An integer-relation search found no integer polynomial of degree at most 8 with coefficients bounded by 10^6 that annihilates C_{31}^4.

7.2. **Evidence that the relaxation is exact.**

Proposition 7.1. *Let $f^\star$ be any zero-mean vector with $F(f^\star) = C_{31}^4 G(f^\star)$, and put $\kappa = \|z(f^\star)\|^2/G(f^\star)$. If (t, Q, C) is feasible for the program of Section 6 with margin γ, then $t \geq C_{31}^4 + \kappa\gamma$.*

Proof. Evaluate the identity at $f^\star$: $(t - C_{31}^4)G(f^\star) = z(f^\star)^{\mathsf{T}} Q z(f^\star) \geq \gamma\|z(f^\star)\|^2$. □

For the numerical maximizer, $\kappa \approx 11.94$. The solver gives $(t - C_{31}^4)/\gamma \approx 12.9$ at $\gamma = 10^{-7}$. At $\gamma = 10^{-8}$ the ratio is ≈ 21, which is within the solver's absolute accuracy of about 10^{-8}. Both values are consistent with $t(\gamma) = C_{31}^4 + O(\gamma)$, that is, with the degree-four relaxation with zero-mean multiplier being exact. Without the multiplier, the same program converges to $t^{1/4} \approx 1.5698391$. This equals the numerical norm of H on all of $\mathbb{R}^{31}$, so that relaxation is also exact, but for the larger, unconstrained problem.

7.3. **Other orders.** Table 2 lists the best values found by multistart BFGS (60 starts) for the analogous zero-mean problem on $\mathbb{Z}/n\mathbb{Z}$ with odd n. These values are not certified.

TABLE 2. Numerical values of the zero-mean L^4 norm of the sign multiplier on $\mathbb{Z}/n\mathbb{Z}$ (not certified).

n	5	7	11	15	21	31	47	63
$C_n \approx$	1.20232	1.29782	1.39908	1.45695	1.51240	1.56938	1.62354	1.65825

The sequence increases slowly. A natural question is how it compares with the Pichorides constant $\cot(\pi/8) = 1 + \sqrt{2}$ for the circle [10].

8. Reproducibility

The supplementary archive contains the certificate, the checker, and all generating code. It is available at `https://github.com/Sergpreneur/paper-p2730`. To verify:

```
python3 -I verify.py certificate.json
```

The checker needs only Python's standard library. On one core it runs in about five minutes, and it prints `VERIFIED` only after all exact assertions pass. As a negative control, the same certificate with U replaced by 1.5693754 is rejected.

The generating scripts (`sdp.py`, `sdp2.py`, `gen.py`, `newton.py`) were run with Python 3.13.16, NumPy 2.5.3, SciPy 1.18.1, CVXPY 1.9.3, Clarabel 0.11.1 and mpmath 1.4.1. The numerical SDP solution is included, so the rationalization step can be rerun without the solver.

9. Concluding remarks

What remains open is to prove that C_{31} equals the critical value found above. If the relaxation is exact, as the evidence of Section 7 suggests, then an exact SOS identity with $\mu = 0$ should exist. Its Gram matrix would be singular and its coefficients would lie in a number field containing the coordinates of the extremizer. Constructing such an identity exactly is substantially harder than the strict, rounded certificate used here. A cheaper intermediate step is to narrow the bracket with an extended-precision SDP solver, since our margin is limited by the accuracy of a double-precision interior-point method. The same method applies verbatim to other odd orders n, with Gram matrices of size $\frac{n+1}{2} \cdot n$.

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